

A1

Part 1 : Shortest Distance from Point to function

a) Analytical Approach

Given: $y = x^2 + 5$ (0,0) (-4,0) (-8,0) (2,0) (6,0)

① $f(x) = x^2 + 5$
 $f'(x) = 2x$

② $D(x) = (x - x_0)^2 + (f(x) - y_0)^2$
 $D'(x) = 2(x - x_0) + 2(f(x) - y_0)f'(x) = 0$

for all points $y_0 = 0$

$$\begin{aligned} 0 &= 2(x - x_0) + 2f(x)f'(x) \\ 0 &= 2[(x - x_0) + f(x)f'(x)] \\ 0 &= (x - x_0) + (x^2 + 5)(2x) \\ 0 &= x - x_0 + 2x^3 + 10x \\ 0 &= 2x^3 + 11x - x_0 \end{aligned}$$

③ sub in $x_0 = -4$ (for point (-4,0))

$0 = 2x^3 + 11x + 4$
 $x^* = -0.35546$ } solve eqn for $x = x^*$

$f(x) = y = x^2 + 5$
 $= 5.12635$

∴ point on curve closest to line is (-0.355, 5.126)

④ Calculate min distance

$$\begin{aligned} d &= \sqrt{(x^* - x_0)^2 + (y^* - y_0)^2} \\ &= \sqrt{(-0.355 + 4)^2 + (5.126 - 0)^2} \\ &= 6.2898 \end{aligned}$$

Summary:

(x_0, y_0)	equation	(x^*, y^*)	dist
(0,0)	$2x^3 + 11x = 0$	(0, 5)	5
(-4,0)	$2x^3 + 11x + 4 = 0$	(-0.355, 5.126)	6.289
(-8,0)	$2x^3 + 11x + 8 = 0$	(-0.672, 5.451)	9.133
(2,0)	$2x^3 + 11x - 2 = 0$	(0.1807, 5.0326)	5.351
(6,0)	$2x^3 + 11x - 6 = 0$	(0.5199, 5.2702)	7.603

b) Numerical Approach

i) Newton-Raphson

$f(x) = x^2 + 5$
 $f'(x) = 2x$

For point (-4,0)

$$\begin{aligned} D(x) &= (x - x_0)^2 + (f(x) - y_0)^2 \\ D'(x) &= 2(x - x_0) + 2(f(x) - y_0)f'(x) \\ &= 2(x + 4) + 2(x^2 + 5 - 0)(2x) \\ &= 2x + 8 + (2x^2 + 10)(2x) \\ &= 2x + 8 + 4x^3 + 20x \\ &= 4x^3 + 22x + 8 \\ D''(x) &= 12x^2 + 22 \end{aligned}$$

$$\begin{aligned} x_{n+1} &= x_n - \frac{D'(x_n)}{D''(x_n)} \\ &= x_n - \frac{4x^3 + 22x + 8}{12x^2 + 22} \end{aligned}$$

Initial guess $x_0 = 0$

Iteration #1:

$x_0 = 0$
 $D'(x) = 8$ $D''(x) = 22$
 $x_{n+1} = 0 - \frac{8}{22}$
 $x_1 = -\frac{8}{22} = -0.3636$

Iteration #2:

$x_1 = -0.3636$
 $D'(x) = -0.1923$ $D''(x) = 23.5867$
 $x_2 = -0.3636 - \frac{-0.1923}{23.5867}$
 $x_2 = -0.3494$

Iteration #3:

$x_2 = -0.3494$
 $D'(x) = 0.14155$ $D''(x) = 23.465$
 $x_3 = -0.3494 - \frac{0.14155}{23.465}$
 $x_3 = -0.35543$

Iteration #4:

$x_3 = -0.35543$
 $D'(x) = 0.000877$ $D''(x) = 23.5157$
 $x_4 = -0.35543 - \frac{0.000877}{23.5157}$
 $x_4 = -0.35543$

∴ $x^* = -0.35543$ $y^* = 5.1255$

$$\begin{aligned} D(x^*) &= \sqrt{(x^* - x_0)^2 + (y^* - y_0)^2} \\ &= \sqrt{(-0.35543 + 4)^2 + (5.1255 - 0)^2} \\ &= 6.28984 \approx d_{min} \end{aligned}$$

Point (0,0):

$$\begin{aligned} D(x) &= 2(x-x_0) + 2(f(x) - y_0) f'(x) \\ &= 2x + 2(x^2 + 5)(2x) \\ &= 2x + 4x^3 + 20x \\ &= 4x^3 + 22x \end{aligned}$$

Iteration #1:

$$\begin{aligned} \text{initial guess } x_0 &= 0 \\ D'(0) &= 0 & D''(0) &= 22 \\ x_1 &= 0 - \frac{0}{22} \\ &= 0 \end{aligned}$$

Iteration #2:

$$\begin{aligned} D(0) &= 0 & D''(0) &= 22 \\ x_2 &= 0 - \frac{0}{22} \\ &= 0 \end{aligned}$$

Point (1,0):

$$\begin{aligned} D(x) &= 2(x-x_0) + 2(f(x) - y_0) f'(x) \\ &= 2(x-2) + 2(x^2 + 5)(2x) \\ &= 2x - 4 + 4x^3 + 20x \\ &= 4x^3 + 22x - 4 \end{aligned}$$

Iteration #1:

$$\begin{aligned} x_0 &= 0 \\ D'(0) &= -4 & D''(0) &= 22 \\ x_1 &= 0 - \left(\frac{-4}{22} \right) \\ &= 2/11 \end{aligned}$$

Iteration #2:

$$\begin{aligned} D(2/11) &= \frac{32}{1331} & D''(2/11) &= \frac{2710}{121} \\ x_2 &= \frac{2}{11} - \frac{32/1331}{2710/121} \\ &= 0.1807447... \end{aligned}$$

2) Golden Section Search

$$D(x) = (x-x_0)^2 + (y-y_0)^2$$

for point (-4,0)

$$\begin{aligned} D(x) &= (x+4)^2 + y^2 \\ &= (x+4)^2 + (x^2 + 5) \end{aligned}$$

interval: $[-1, 0] \leftarrow [a, b]$

$$\begin{aligned} D(-1) &= (3)^2 + (1+5)^2 = 49 \\ D(-0.5) &= (3.5)^2 + (5.25)^2 = 39.8125 \\ D(0) &= 4^2 + 5^2 = 41 \end{aligned}$$

↑
this is less than end points
so includes the minimum

$$\begin{aligned} \phi &= \frac{1+\sqrt{5}}{2} & r &= 2-\phi \\ & & r &= 2 - \left(\frac{1+\sqrt{5}}{2} \right) \\ & & r &= \frac{3-\sqrt{5}}{2} \end{aligned}$$

Iteration #1:

$$\begin{aligned} x_1 &= a + r(b-a) & x_2 &= b - r(b-a) \\ &= -1 + \left(\frac{3-\sqrt{5}}{2} \right) (0+1) & &= 0 - \left(\frac{3-\sqrt{5}}{2} \right) (0+1) \\ &= \frac{1-\sqrt{5}}{2} \approx -0.618034 & &= \frac{\sqrt{5}-3}{2} \approx -0.381966 \\ D(x_1) &= 40.40325 & D(x_2) &= 39.570436 \end{aligned}$$

$$\begin{aligned} D(x_2) &< D(x_1) \\ \text{New interval} &: [x_1, b] \end{aligned}$$

Iteration #5: $[-0.472135, -0.32673]$

$$\begin{aligned} x_1 &= -0.416404 & x_2 &= -0.381967 \\ D(x_1) &= 39.60614 & D(x_2) &= 39.57043 \end{aligned}$$

$$\begin{aligned} x^* &= \text{midpoint bwn } x_1 \text{ \& } x_2 \\ x^* &= \frac{-0.416404 + (-0.381967)}{2} = -0.39918 \end{aligned}$$

$$D(x^*) = 39.58474$$

$$d_{\min} = \sqrt{D(x^*)} = 6.29164...$$

Point (-8,0):

$$\begin{aligned} D(x) &= 2(x-x_0) + 2(f(x) - y_0) f'(x) \\ &= 2(x+8) + 2(x^2 + 5)(2x) \\ &= 2x + 16 + 4x^3 + 20x \\ &= 4x^3 + 22x + 16 \end{aligned}$$

Iteration #1
 $x_0 = 0$

$$\begin{aligned} D'(0) &= 16 & D''(0) &= 22 \\ x_1 &= 0 - \frac{16}{22} \\ &= 8/11 \end{aligned}$$

Iteration #2

$$\begin{aligned} D'(8/11) &= 33.5386 & D''(8/11) &= 28.3471 \\ x_2 &= \frac{8}{11} - \frac{33.5386}{28.3471} \\ &= -0.45586 \end{aligned}$$

Point (6,0):

$$\begin{aligned} D(x) &= 2(x-x_0) + 2(f(x) - y_0) f'(x) \\ &= 2(x-6) + 2(x^2 + 5)(2x) \\ &= 2x - 12 + 4x^3 + 20x \\ &= 4x^3 + 22x - 12 \end{aligned}$$

Iteration #1

$$\begin{aligned} x_0 &= 0 \\ D'(0) &= -12 & D''(0) &= 22 \\ x_1 &= 0 - \left(\frac{-12}{22} \right) \\ &= 6/11 \end{aligned}$$

Iteration #2

$$\begin{aligned} D'(6/11) &= 0.649135 & D''(6/11) &= \frac{2094}{121} \\ x_2 &= \frac{6}{11} - \frac{0.649135}{2094/121} \\ &= 0.52006 \end{aligned}$$

Iteration #2: $[-0.618034, 0]$

$$\begin{aligned} x_1 &= -0.618034 + \left(\frac{3-\sqrt{5}}{2} \right) (0.618034) & x_2 &= 0 - \left(\frac{3-\sqrt{5}}{2} \right) (0.618034) \\ &= -0.381966 & &= -0.236068 \end{aligned}$$

$$D(x_1) = 39.12243$$

$$D(x_2) = 39.12757$$

$$D(x_2) < D(x_1)$$

New interval: $[x_2, x_1]$

Iteration #4: $[-0.472135, -0.326068]$

$$\begin{aligned} x_1 &= -0.381965 & x_2 &= -0.32623 \\ D(x_1) &= 39.5704 & D(x_2) &= 39.57167 \end{aligned}$$

$$D(x_2) > D(x_1)$$

New interval: $[a, x_2]$

Iteration #3: $[-0.618034, -0.236068]$

$$\begin{aligned} x_1 &= -0.472135 & x_2 &= -0.381966 \\ D(x_1) &= 39.72464 & D(x_2) &= 39.5704 \end{aligned}$$

$$D(x_2) < D(x_1)$$

New interval: $[x_1, b]$

Point (0,0):

$$D(x) = x^2 + (x^2 + 5)^2$$

Interval: $[-1, 0.5]$

Iteration #1:

$$\begin{aligned} x_1 &= a + r(b-a) \\ &= -1 + \left(\frac{3-\sqrt{5}}{2}\right)(0.5 - (-1)) \\ &= \frac{5-3\sqrt{5}}{4} \approx -0.42705 \end{aligned}$$

$$D(x_1) = 27.0393$$

$$D(x_1) > D(x_2)$$

New interval $[x_1, b]$

$$\begin{aligned} x_2 &= b - r(b-a) \\ &= 0.5 - \left(\frac{3-\sqrt{5}}{2}\right)(0.5 + 1) \\ &= \frac{-7+3\sqrt{5}}{4} \approx -0.07295 \end{aligned}$$

$$D(x_2) = 25.0585$$

Iteration #2:

$[-0.42705, 0.5]$

$$x_1 = \frac{-13+5\sqrt{5}}{4} \approx -0.4549 \quad x_2 = \frac{7-3\sqrt{5}}{2} \approx 0.145898$$

$$D(x_1) = 27.3192$$

$$D(x_2) = 25.2346$$

$$D(x_1) > D(x_2)$$

new interval $[x_1, b]$

Point (-8,0):

$$D(x) = (x+8)^2 + (x^2+5)^2$$

Iteration #1: $[-1, 0]$

$$x_1 = \frac{1-\sqrt{5}}{2} \approx -0.618034 \quad x_2 = \frac{\sqrt{5}-3}{2} \approx -0.381966$$

$$D(x_1) = 83.4589$$

$$D(x_2) = 84.5147$$

$$D(x_1) < D(x_2)$$

$\Rightarrow [a, x_2]$

Iteration #2: $[-1, -0.381966]$

$$x_1 = -3+\sqrt{5} \approx -0.76393 \quad x_2 = \frac{1-\sqrt{5}}{2} \approx -0.61803$$

$$D(x_1) = 83.537$$

$$D(x_2) = 83.4589$$

$$D(x_1) > D(x_2)$$

$\Rightarrow [x_1, b]$

Point (2,0)

$$D(x) = (x-2)^2 + (x^2+5)^2$$

Iteration #1: $[0, 1]$

$$x_1 = \frac{3-\sqrt{5}}{2} \approx 0.381966 \quad x_2 = \frac{-1+\sqrt{5}}{2} \approx 0.61803$$

$$D(x_1) = 29.0983$$

$$D(x_2) = 51.9455 \approx 30.875$$

$$D(x_1) < D(x_2)$$

$\Rightarrow [a, x_2]$

Iteration #2: $[0, 0.61803]$

$$x_1 = -2+\sqrt{5} \approx 0.23607 \quad x_2 = \frac{3-\sqrt{5}}{2} \approx 0.38196$$

$$D(x_1) = 28.6718$$

$$D(x_2) = 29.098$$

$$D(x_1) < D(x_2)$$

$\Rightarrow [a, x_2]$

Point (6,0)

$$D(x) = (x-6)^2 + (x^2+5)^2$$

Iteration #1: $[0, 1]$

$$x_1 = \frac{3-\sqrt{5}}{2} \approx 0.381966 \quad x_2 = \frac{-1+\sqrt{5}}{2} \approx 0.61803$$

$$D(x_1) = 58.0426$$

$$D(x_2) = 87-13\sqrt{5} \approx 57.9311$$

$$D(x_1) > D(x_2)$$

$\Rightarrow [x_1, b]$

Iteration #2: $[0.38196, 1]$

$$x_1 = \frac{-1+\sqrt{5}}{2} \approx 0.61803 \quad x_2 = 3-\sqrt{5} \approx 0.7639$$

$$D(x_1) = 87-13\sqrt{5} \approx 57.9311$$

$$D(x_2) = 58.5929$$

$$D(x_1) < D(x_2)$$

$\Rightarrow [a, x_2]$

Part 2 : Fitting to an Equation

a) Analytical Approach

Given points (0,0.5) (2,3.5) (1,1.5) (3,7.5)

x_i	y_i	
0	0.5	$n=4$
1	1.5	$\sum x_i = 6$
2	3.5	$\sum x_i^2 = 14$
3	7.5	$\sum y_i = 13$
		$\sum y_i^2 = 71$

1) LINE: $\hat{y}_i = mx + b$

$$e_i = \hat{y}_i - y_i$$

divide by 4 bc 4 data points

$$\begin{aligned} \text{MSE}(m, b) &= \frac{1}{4} \sum_{i=1}^4 (mx_i + b - y_i)^2 \\ &= \frac{1}{4} [(m(0)+b-0.5)^2 + (m(1)+b-1.5)^2 + (m(2)+b-3.5)^2 + (m(3)+b-7.5)^2] \\ &= \frac{1}{4} [(b-0.5)^2 + (m+b-1.5)^2 + (2m+b-3.5)^2 + (3m+b-7.5)^2] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \sum_{i=1}^4 (mx_i + b - y_i)(mx_i + b - y_i) \\
&= \frac{1}{4} \sum_{i=1}^4 m^2 x_i^2 + mbx_i - mx_i y_i + mbx_i + b^2 - by_i - mx_i y_i - by_i + y_i^2 \\
&= \frac{1}{4} [14m^2 + 6mb - 31m + 6mb + 4b^2 - 13b - 31m - 13b + 71] \\
&= \frac{1}{4} [14m^2 + 12mb - 62m + 4b^2 - 26b + 71] \\
&= 3.5m^2 + 3mb - 15.5m + b^2 - 6.5b + 17.75
\end{aligned}$$

$$\begin{aligned}
&\text{MSE:} \\
&\text{SSE} = (-0.2 - 0.5)^2 + (2.1 - 1.5)^2 + (4.4 - 3.5)^2 + (6.7 - 7.5)^2 \\
&= 2.3 \\
&\text{MSE} = \frac{\text{SSE}}{4} = 0.575
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \text{MSE}}{\partial m} &= 7m + 3b - 15.5 = 0 \\
7m + 3b &= 15.5 \\
14m + 6b &= 31 \\
\frac{14m + 6b}{6m + 4b} &= \frac{31}{19.5} \\
5m &= 11.5 \\
m &= 2.3 \\
\therefore y &= 2.3x - 0.2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \text{MSE}}{\partial b} &= 3m + 2b - 6.5 = 0 \\
3m + 2b &= 6.5 \\
6m + 4b &= 13 \\
9m + 6b &= 19.5 \\
b &= \frac{19.5 - 9m}{6} \\
&= \frac{19.5 - 9(2.3)}{6} \\
&= -0.2
\end{aligned}$$

2) PARABOLA : $\hat{y} = ax^2 + bx + c$

x_i	y_i	$\sum x_i^2 = 14$	$\sum x_i^3 = 36$	$\sum x_i y_i = 31$	$\sum x_i^4 = 98$	$\sum x_i^2 y_i = 83$	$\sum y_i = 13$
0	0.5						
1	1.5						
2	3.5						
3	7.5						

$$\begin{aligned}
\theta &= y_i - \hat{y}_i \\
\text{MSE}(a, b, c) &= \frac{1}{4} \sum_{i=1}^4 (ax_i^2 + bx_i + c - y_i)^2 \\
&= \frac{1}{4} \sum_{i=1}^4 (ax_i^2 + bx_i + c - y_i)(ax_i^2 + bx_i + c - y_i) \\
&= \frac{1}{4} \sum_{i=1}^4 a^2 x_i^4 + a^2 x_i^3 + ac x_i^2 - ax_i^3 y_i + ab x_i^3 + b^2 x_i^2 + bc x_i - bx_i y_i + ac x_i^2 + bc x_i + c^2 - cy_i \\
&\quad - ax_i^2 y_i - bx_i y_i - cy_i + y_i^2 \\
&= \frac{1}{4} \sum_{i=1}^4 a^2 x_i^4 + 2ab x_i^3 + 2ac x_i^2 - 2a x_i^3 y_i + b^2 x_i^2 + 2bc x_i - 2b x_i y_i + c^2 - 2c y_i + y_i^2 \\
&= \frac{1}{4} [98a^2 + 72ab + 28ac - 166a + 14b^2 + 12bc - 62b + 4c^2 - 26c + 71] \\
&= 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \text{MSE}}{\partial a} &= 49a + 18b + 7c - 41.5 = 0 \\
98a + 36b + 14c &= 83 \\
98a + 36(2.3 - 3a) + 14\left(\frac{13 - 14a - 6b}{4}\right) &= 83 \\
98a + 82.8 - 108a + 45.5 - 49a - 21b &= 83 \\
-59a - 21b &= -45.3 \\
45.3 &= 59a + 21(2.3 - 3a) \\
45.3 &= 59a + 48.3 - 63a \\
4a &= 3 \\
a &= 0.75 \\
\frac{\partial \text{MSE}}{\partial b} &= 7b + 18a + 3c - 15.5 = 0 \\
36a + 14b + 6c &= 31 \\
36a + 14b + 6\left(\frac{13 - 14a - 6b}{4}\right) &= 31 \\
36a + 14b + 19.5 - 21a - 9b &= 31 \\
15a + 5b &= 11.5 \\
b &= \frac{11.5 - 15a}{5} \\
b &= 2.3 - 3a \\
b &= 2.3 - 3(0.75) \\
&= 0.05 \\
\frac{\partial \text{MSE}}{\partial c} &= 2c + 7a + 3b - 6.5 = 0 \\
14a + 6b + 4c &= 13 \\
c &= \frac{13 - 14a - 6b}{4} \\
c &= \frac{13 - 14(0.75) - 6(0.05)}{4} \\
&= 0.55 \\
\therefore \hat{y} &= 0.75x^2 + 0.05x + 0.55
\end{aligned}$$

$$\begin{aligned}
&\text{MSE:} \\
&\text{SSE} = ((0.55 - 0.5)^2 + (1.35 - 1.5)^2 + (3.65 - 3.5)^2 + (7.45 - 7.5)^2) \\
&= (0.05)
\end{aligned}$$

$$\text{MSE} = \frac{\text{SSE}}{4} = 0.0125$$

b) Numerical Approach - Newton Raphson Method

1) LINE : $\hat{y} = mx + b$

$$MSE(m, b) = 3.5m^2 + 3mb - 15.5m + b^2 - 6.5b + 17.5$$

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5$$

$$\frac{\partial^2 MSE}{\partial m^2} = 7$$

$$\frac{\partial^2 MSE}{\partial b^2} = 2$$

① set :

$$m_{n+1} = m_n - \frac{7m_n + 3b - 15.5}{7}$$

$$b_{n+1} = b_n - \frac{3m_{n+1} + 2b_n - 6.5}{2}$$

$$= 0 - \frac{7(0) + 3b - 15.5}{7}$$

$$= 0 - \left(\frac{3m_{n+1} - 6.5}{2} \right)$$

$$= \frac{15.5 - 3b}{7}$$

$$= \frac{6.5 - 3m_{n+1}}{2}$$

② choose initial values : $m_0 = 0$ $b_0 = 0$

Iteration #1

$$m_1 = \frac{15.5 - 3(0)}{7}$$

$$b_1 = \frac{6.5 - 3m_1}{2}$$

$$= 2.214285..$$

$$= \frac{6.5 - 3(2.214285..)}{2}$$

$$= -0.0714285..$$

$$\hat{y} = 2.24x - 0.0714$$

$$SSE = (-0.0714 - 0.05)^2 + (2.1426 - 1.5)^2 + (4.3566 - 3.5)^2 + (6.5706 - 7.5)^2$$

$$= 2.02522$$

$$MSE = \frac{SSE}{4} = 0.50630..$$

2) PARABOLA : $\hat{y} = ax^2 + bx + c$

$$MSE(a, b, c) = 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75$$

$$\frac{\partial MSE}{\partial a} = 49a + 18b + 7c - 41.5$$

$$\frac{\partial MSE}{\partial b} = 7b + 18a + 3c - 15.5$$

$$\frac{\partial MSE}{\partial c} = 2c + 7a + 3b - 6.5$$

$$\frac{\partial^2 MSE}{\partial a^2} = 49$$

$$\frac{\partial^2 MSE}{\partial b^2} = 7$$

$$\frac{\partial^2 MSE}{\partial c^2} = 2$$

$$a_{n+1} = a_n - \frac{49a_n + 18b_n + 7c_n - 41.5}{49}$$

$$= \frac{41.5 - 18b_n - 7c_n}{49}$$

$$b_{n+1} = b_n - \frac{7b_n + 18a_n + 3c_n - 15.5}{7}$$

$$= \frac{15.5 - 18a_{n+1} - 3c_n}{7}$$

$$c_{n+1} = c_n - \frac{2c_n + 7a_n + 3b_n - 6.5}{2}$$

$$= \frac{6.5 - 7a_{n+1} - 3b_{n+1}}{2}$$

③ initial params :

$$a_0 = 0 \quad b_0 = 0 \quad c_0 = 0 \Rightarrow \hat{y} = 0$$

Iteration #1 :

$$a_1 = \frac{41.5 - 18(0) - 7(0)}{49}$$

$$= 0.8469387..$$

$$b_1 = \frac{15.5 - 18(0.8469) - 3(0)}{7}$$

$$= 0.036443..$$

$$c_1 = \frac{6.5 - 7(0.8469) - 3(0.036443)}{2}$$

$$= 0.231185..$$

$$\therefore \hat{y} = 0.847x^2 + 0.036x + 0.231$$

MSE :

$$SSE = (0.231 - 0.5)^2 + (1.114 - 1.5)^2 + (3.691 - 3.5)^2 + (7.962 - 7.5)^2$$

$$= 0.450281$$

$$MSE = \frac{SSE}{4} = 0.11257$$

③

Iteration #2

$$m_2 = \frac{15.5 - 3b_1}{7}$$

$$b_2 = \frac{6.5 - 3(2.244898)}{2}$$

$$= \frac{15.5 - 3(-0.0714285)}{7}$$

$$= -0.1734$$

$$= 2.244898$$

$$\hat{y} = 2.24x - 0.117$$

$$SSE = (-0.117 - 0.5)^2 + (2.123 - 1.5)^2 + (4.363 - 3.5)^2 + (6.603 - 7.5)^2$$

$$= 2.318196$$

$$MSE = \frac{SSE}{4} = 0.579549..$$

Iteration #2 :

$$a_1 = \frac{41.5 - 18(0.036) - 7(0.231)}{49}$$

$$= 0.800714$$

$$b_1 = \frac{15.5 - 18(0.8007) - 3(0.231)}{7}$$

$$= 0.056342$$

$$c_1 = \frac{6.5 - 7(0.800714) - 3(0.056342)}{2}$$

$$= 0.263012..$$

$$\hat{y} = 0.801x^2 + 0.056x + 0.263$$

MSE :

$$SSE = (0.263 - 0.5)^2 + (1.22 - 1.5)^2 + (3.679 - 3.5)^2 + (7.74 - 7.5)^2$$

$$= 0.19681$$

$$MSE = \frac{SSE}{4} = 0.0492$$